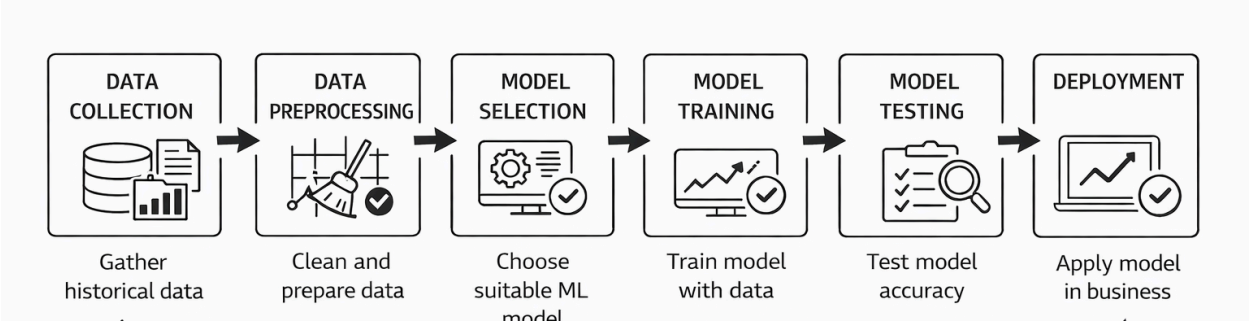
UNIT III – BUSINESS FORECASTING

MACHINE LEARNING FOR PREDICTIVE ANALYTICS

Machine Learning (ML) in predictive analytics is the use of **algorithms and data** to **predict future outcomes** such as sales, demand, or trends based on past data.

PROCESS OF ML IN PREDICTIVE ANALYTICS



STEPS INVOLVED

1. Data Collection

- Collect historical data (sales, customer data, market trends)
- Data can be structured or unstructured

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2. Data Preprocessing

- Clean data (remove errors, missing values)
- Convert data into usable format

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3. Model Selection

- Choose suitable ML model
- Example: Linear Regression, Decision Tree

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4. Model Training

- Train model using past data
- Model learns patterns and relationships

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5. Model Testing

- Test model accuracy using new data
- Check performance

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6. Prediction

- Use model to predict future outcomes
- Example: future sales, demand

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7. Deployment

- Apply model in real business environment
- Support decision making

TYPES OF MACHINE LEARNING USED

1. Supervised Learning

- Uses labeled data
- Example: Sales prediction

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2. Unsupervised Learning

- Finds patterns without labeled data
- Example: Customer segmentation

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3. Reinforcement Learning

- Learns through trial and error
- Example: Recommendation systems

COMMON ML ALGORITHMS

- Linear Regression → Predict numerical values
- Decision Tree → Classification and prediction
- Random Forest → Improved accuracy
- K-Means Clustering → Group customers
- Neural Networks → Complex predictions

APPLICATIONS IN BUSINESS FORECASTING

- Sales forecasting
- Demand prediction
- Customer behaviour prediction
- Risk analysis
- Inventory planning

ADVANTAGES

- High accuracy
- Handles large data
- Improves decision making
- Saves time and cost